

variables

top
bot - bottom of symbol table
blk1 - points to buffer last
list - bottom of symbol table
memory - next storage location
factor - relocation factor
guess -
blk
table - dispatch table for type of block
last - 104 words buffer at end of loader
blk2
first
blk3
checksum - Checksum
blk4 - list
value - cumulative value of symbols already processed
bitvec - counter for relocation bit pairs
blkj - holds address of block type dispatch table
blk - jump
codes - storage for relocation bits word

7775 - holds entry to symbol package punch
7776 - holds entry jump to loader
7777 -

subroutines

bits - get next two relocation bits + process with word
0 - Take care of external symbol reenter bits from top
others get value of
getword - get a word

ad - tail

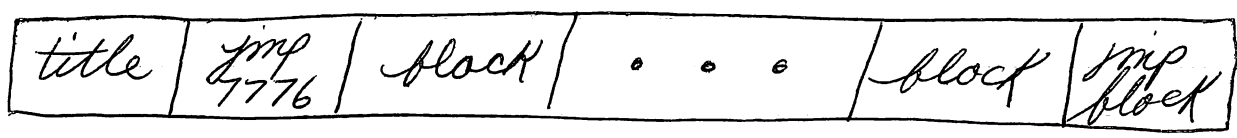
XSym - calculates value of symbol if available

Operation of Linking Loader

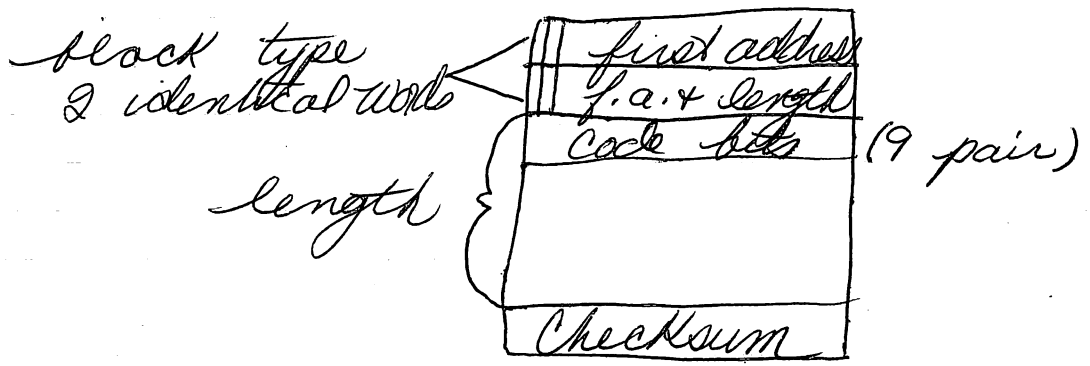
SS 1 ↓ print out starting position
2 ↑ read starting location from TW
3
4
5
6

Linking Loader

General Tape Format



General Block Format



Block Types

Absolute - 0

load at address specified
(note there may be relocatable words in an absolute block - references to a non-absolute block)

Relocatable - 1

take relocation factor, add address, load there

Jump - 2

one word of data which is the starting address of the program just loaded
format is the same as above
2 blocks

Block Types, Cont.

②

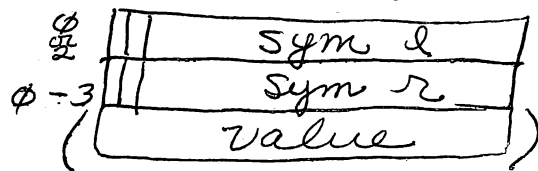
Library - 3

Simply a list of global symbols
if any one is needed the next
data block are loaded. if
not, blocks are skipped until
another library or a jump block

Code Bits

External Symbol - 0

Read next two words as
symbol (Sgoys code)



Sym 2 always exists
Sym 1 may not

Sym 2 code bits

Look up in Symbol table - 0

if Sym 1 = 0 add value
from table to current
word

if Sym 1 = 2 subtract value
in table from current
word

Use absolute value - 1

read next word
as value and add
factor masked to
1:2 bits if Sym 1
= 0; 16 bits if = 2

Relocate subtract - 2

Relocate add - 3

if a symbol is not in the
Symbol table - add it to the
table with the current memory
address and whether to add
or subtract

if in but not defined add
current address to list of places
to update

if in and currently being defined
update according to list

if defined and being defined, complain
if the two values don't match

Don't Relocate - 1

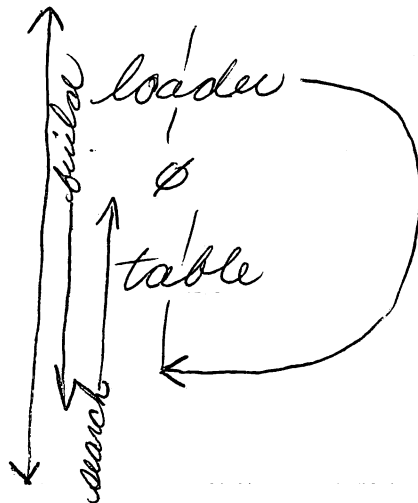
Relocate subtract - 2

Relocate - 3

Take next full word, add factor,
put in memory

Symbol Table Structure

7777



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Symbol Table Structure, cont.

④

Defined Entry

value
(sym l)

0	Sym r
---	-------

└ 0 Sym l exists
1 Sym l doesn't exist

Undefined Entry

address n

.

.

.

address 1

n

(sym l)

1	Sym r
---	-------

└ 0 sym l exists
1 Sym l doesn't exist